



for its place of use. Good product design needs to carefully evaluate what features to provide in the product to make it most useful.

Let us take designing of windmill charge controller system as an example to understand the concept. I have selected this example on purpose. First of all, the device is rather simple, which will allow us to concentrate on the design process rather than getting involved with electronic circuits. The second reason for selecting the charge controller is that it will be functioning somewhere in obscurity and forgotten. We shall try to change that and explore ways to make the charge controller a lovable item.

Wind turbines are mechanical systems. If we simply disconnect the battery, once it is fully charged, the turbine will start rotating rapidly in absence of any load. Such rapid rotation may lead to turbine failure. When the turbine produces more power than required, the extra power needs to be diverted to auxiliary equipment like some pump or decorative lighting, etc. The charge controller does this work.

The windmill charge controller has multiple roles to play. It prevents overcharging of the battery, ensures that charge does not flow back from the battery to the turbine, and it also needs to protect the turbine from rotating too rapidly. Windmills generate alternating current, which is rectified before use.

Charge controller sits in between the rectifier and battery. A dummy load is attached to the charge controller (see Fig. 2). In case the turbine starts rotating too rapidly, the charge controller activates a braking switch to slow down the rotation.

In this basic configuration, customers do not interact with the charge controller directly. Such an item will be treated as an appendage of the windmill and treated as a commodity item. Such a product

will have multiple competitors. Each of them will try to compete on cost and reliability.

Unless you produce the controllers in huge quantity it is very difficult to compete on these two parameters alone. There is no way you, as a designer, can make such a product stand out from the others in competition.

The charge controller performs some useful functions but it falls short on the frequency of use. Our innovation to make the charge controller a lovable product has to be directed towards increasing the frequency of use.

The basic function of charging a battery or diverting the power to a dummy load does not have any user interaction. We need to think of adding additional functions to the charge controller to do that. If we consider that the essential function of a charge controller is to protect the battery then adding battery status monitoring is something that we can provide easily.

Other parameters that the charge controller monitors is the power generated and power consumed. Keeping track of these parameters can add value to the homeowner. If the user knows the power generation and consumption patterns then he or she can plan the load distribution more intelligently. These parameters can also tell us about the battery condition and warn us of any deterioration of battery capacity.

It is easy to see that a smart charge controller that gives us all this information will bring a product from obscurity to the centre of attention. We have various alternatives for making the charge controller share this information. This information can be stored internally and analysed using an onboard computer. Another alternative can be storing this information in a cloud server where the analysis can be done. Third option may be to store

the information only and transfer the data into a portable memory card, which in turn can be analysed on a desktop or laptop computer.

Each of these options will have different design and cost implications. Product performance and technical features like power requirements, voltage, and current are important as the controller must operate in conjunction with readily available products being sold in the market.

While deciding on the final product specifications we need to look at the market competition, the features being offered by other manufacturers as well as their prices. After all, a lovable product will also need to be affordable. For developing the product we need to consider many other factors like the environment it will operate in, various governmental and other regulations, besides making provisions for the future requirements that may arise.

When we start to define a product, we shall have many options and shall decide on certain options. At the product definition stage, these decisions do not cost much effort or money but these have far-reaching consequences.

A study done on cost commitment at various stages of design found that about 63% of product cost gets frozen by the end of conceptual design. By the time a product design is made more than 80% of the product cost is committed. Improvements and innovation done in the subsequent stages are only limited to the remaining 20% of the cost.

Hence, a lot of care should be taken in deciding the correct product specifications and the correct design options. These analyses lay the foundation for the rest of the product development process. In our next article, we shall discuss the process of arriving at the product specification systematically. **EFY**